

Q-1 What is Mobile Computing?

→ Mobile computing refers to ability to use computing devices while being mobile and connected to a network. It allows users to transmit data, voice, and video through wireless-enabled devices without the need for a fixed physical connection. Mobile computing involves portable devices like smartphones, laptops, and tablets that can access data and applications through wireless networks such as Wi-Fi, cellular (3G/4G/5G), and Bluetooth. It offers flexibility, portability, and real-time communication, enabling users to work or access services from any location. Key features include mobility, broad network accessibility, and context awareness. It is widely used in fields like business, healthcare, education, and logistics.

Q-2 Which of the following is NOT an application of Mobile Computing?

→ Not all applications are suitable for mobile computing. An application that does not support mobility, relies on constant power supply, or needs high-end computing hardware (like CAD software or some desktop-only enterprise applications)

is not a mobile computing application. These apps usually require a stationary setup and cannot run on mobile platforms due to hardware/software limitations.

Q-3 What is a limitation of Mobile Computing?

-> While mobile computing offers many benefits, it also comes with limitations. The most common limitations include battery constraints, limited storage and processing power compared to desktops, and dependency on wireless networks. Poor or unstable network connectivity can interrupt access to cloud services or real-time communication. Security is another concern, as mobile devices are more vulnerable to theft and data breaches.

Q-4 What does the architecture of Mobile Computing typically include?

-> Mobile computing architecture consists of three main components: mobile devices (clients), communication networks, and backend infrastructure (servers/cloud).

- Mobile Devices: Laptops, smartphones, and tablets used by end-users.

- **Wireless Networks:** Enable connectivity using protocols like 3G/4G/5G, Wi-Fi, Bluetooth.
- **Middleware:** Software that bridges communication between the device and backend systems.

Q-5 What is the primary function of Cellular Networks?

→ Cellular networks provide wireless communication services that allow users to make voice calls, send text messages, and access the Internet from mobile devices. They are based on the division of a geographic area into cells, each served by a cell tower. Cellular networks support mobility by allowing users to move across different cells without losing connectivity, using handoff techniques. These networks operate on different generations such as 2G (basic data), 3G (faster data), 4G (high-speed data), and 5G (ultra-fast, low-latency communication). The main function is to provide seamless mobile communication over a wide area.

Q-6 What is Mobile IP used for in Mobile Computing?

→ Mobile IP allows mobile devices to move from one network to another while maintaining a

permanent IP address. This enables uninterrupted communication and session continuity. Without Mobile IP, a change in location would mean a change in the IP address, breaking any ongoing network session. Mobile IP introduces three key entities: Home Agent, Foreign Agent, and Care-of Address. It ensures that all incoming data is rerouted to the device's new location through tunneling. This is essential for applications like VoIP and video streaming where connection stability is critical.

Q-7 Which organization is responsible for the Android platform?

-> Google is the main organization behind the development and maintenance of the Android platform. Android was initially developed by Android Inc., which was acquired by Google in 2005. Google leads the Android Open Source Project (AOSP), which provides the base Android code to manufacturers and developers. While the core Android OS is open source, Google also provides proprietary services like the Play Store, Google Maps, and Google Play Services that come bundled with most Android devices. Google continuously updates Android with new features, security patches, and performance improvements.

Q-8 What does the Android SDK provide?

-> The android Software Development Kit (SDK) provides the tools and libraries necessary to develop Android applications. It includes:

- Android Studio IDE (Official IDE for development)
- Emulator (to test apps without real devices)
- Platform APIs (to access device features)
- Debugging tools (Logcat, ADB)
- Build tools (Gradle)
- UI design tools (layout editor)

It enables developers to write, test, and deploy Android apps efficiently. SDK updates are released with each new Android version to support new APIs and features.

Q-9 Which language is primarily used for Android development?

-> Java has traditionally been the main programming language for Android development. It is supported by the Android SDK and widely used due to its

robust libraries and community support. However, in recent years, Kotlin has become an official Android development language, preferred for its concise syntax, null-safety, and better interoperability with Java. Kotlin is now recommended by Google for new Android projects, although both languages can coexist in the same project. Other languages like C++ (via NDK) are used for native-level coding.

Q-10 What is the purpose of the Android Emulator?

→ The Android Emulator is a virtual device that mimics the functionality of a physical Android device. It allows developers to test applications across various device configurations, Android versions, screen sizes, and hardware features without needing real devices. The emulator is part of Android Studio and supports debugging, simulating phone calls, GPS location, battery states, and more. It helps in faster development and testing, especially in early stages or when devices are unavailable. However, performance can be slower than real devices, particularly without hardware acceleration.

Q-11 What is the main purpose of an Activity in an Android application?

-> In Android, an activity represents a single screen with a user interface, similar to a window or form in desktop applications. It is one of the four core components of Android applications. Each Activity is responsible for displaying UI elements and handling user interactions on a particular screen. For example, a login screen, home page, or settings page is usually an individual activity. Activities follow a lifecycle (`onCreate()`, `onStart()`, `onResume()`, etc.) that Android manages as the user navigates through the app. This lifecycle allows developers to manage resources efficiently. One application can have multiple activities, and navigation between them can be done using Intents.

Q-12 Which file contains essential information about an Android application, like permissions and components?

-> The file that contains all essential metadata about an Android app is `AndroidManifest.xml`. This file is located at the root of the project and plays a crucial role in application configuration. It declares all the app components (like internet access, camera, location), hardware and software requirements, app theme, and minimum SDK version.

Q-13 What is an Intent in Android?

-> An Intent in Android is a messaging object used to request an action from another app component. It is used to start activities, services, or send broadcasts. Intents can be explicit or implicit.

Q-14 Which of the following is NOT a component of an Android application?

-> The four main components of an Android application are:

- Activity - UI screen
- Service - background tasks
- BroadcastReceiver - listens for system/app broadcasts
- ContentProvider - manages shared app data.

Q-15 What is the role of the Application Context in Android?

-> The Application Context is a global context tied to the lifecycle of the application, and it is used when you need access to application-wide resources or services.